

The four-cycle contradicts the minimality claim in Conjecture 3461

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The conjecture includes the claim that the smallest planar graph for which $\chi_{\text{DP}} - \chi = 1$ is a three-dimensional prism. The four-cycle already has this difference and has only four vertices. A prism over a genuine m -gon has $2m \geq 6$ vertices, so the asserted minimality is false.

Let C_4 have successive vertices $0, 1, 2, 3$ and edges $01, 12, 23, 30$. A DP cover assigns a color fiber to each vertex, makes each fiber a clique, and places a matching between fibers of each graph edge. There are no cross-fiber edges over a nonedge. A DP coloring is an independent transversal, choosing one color from every fiber. These are the standard definitions; see Bernshteyn, Kostochka, and Pron, On DP-coloring of graphs and multigraphs.

Theorem 1. C_4 is planar, $\chi(C_4) = 2$, and $\chi_{\text{DP}}(C_4) = 3$.

Proof. The usual alternating coloring of the cycle uses two colors. An edge precludes a one-coloring, so the ordinary chromatic number is two.

To rule out DP two-colorability, give every vertex the fiber $\{0, 1\}$. On edges $01, 12, 23$, match equal colors. On edge 30 , instead match opposite colors. Avoiding the first three matchings forces successive chosen colors to alternate, so vertices 0 and 3 receive opposite colors. Their chosen pair is then forbidden by the last matching. Thus this genuine two-fold cover has no independent transversal.

For the upper bound, take an arbitrary cover with at least three colors available at each vertex, and restrict each fiber to any three of them. Choose a color at vertex 0 . At vertex 1 at most one color is forbidden by the previous choice, so choose another. At vertex 2 again at most one is forbidden. At vertex 3 its two already colored neighbors forbid at most two colors in total. One of its three colors therefore remains. This constructs an independent transversal for every cover, proving DP three-colorability. Covers with empty fibers cannot have a transversal, and a one-fold cover with its edge matchings present also has none; hence the exact DP chromatic number is three.

Finally place the vertices at $(0, 0), (1, 0), (1, 1), (0, 1)$ in that order. Draw each edge as the corresponding side of the unit square. Its sides have disjoint interiors and meet only at the prescribed shared endpoints, so this is a planar embedding. $\square$

The difference is $3 - 2 = 1$ on four vertices. This is smaller than any prism over a polygon; the four-cycle also has fewer edges. The counterexample targets the explicit minimality clause, and makes no claim to disprove the other planar DP-coloring bounds in the source. In particular no girth five hypothesis is imposed on the source's separate difference-spectrum and minimality clause.

Formal verification of covers and chromatic numbers. The Lean graph has the exact cyclic adjacency relation, with symmetry and absence of loops checked. Its cover structure uses arbitrary vertex-dependent color types and arbitrary symmetric, supported matchings; the matchings may be partial. The auxiliary graph on the disjoint union of fibers is defined with the required fiber cliques and cross-fiber edges. The program proves that avoiding all forbidden pairs is equivalent to an independent transversal in this graph.

The twisted two-fold cover satisfies every cover axiom. All 16 possible color selections are checked, and a proved tuple expansion shows that this checks every function on the four vertices. The upper-bound proof is instead symbolic in an arbitrary cover. A small three-color lemma proves that two sets each containing at most one forbidden color cannot exclude all three choices. Restriction from arbitrary color fibers to any three distinct colors is explicitly formalized, with its injectivity and matching bridge. Consequently the upper bound covers all lists of size at least three, not only one selected cover or uniform named color lists. Both chromatic parameters are proved minimal over every smaller natural number of colors.

Formal verification of the drawing and final contradiction. The square certificate defines the vertex positions, all four graph-edge endpoint pairs, and parametrized edge maps $(t, 0), (1, t), (t, 1), (0, t)$. Endpoint incidence, injectivity of vertices and edge maps, exclusion of other vertices from edge interiors, and disjointness of distinct edge interiors are all proved. The proofs quantify over arbitrary scalar types, distinct endpoint coordinates, and arbitrary interior parameters. In particular they apply to every real $t, u \in (0, 1)$; the proof is not an integer-coordinate sampling test. Over the reals these explicit maps are the ordinary continuous straight segments, furnishing the standard planar drawing.

Finally the program proves that $2m > 4$ whenever $m \geq 3$, and negates the necessary instance of the proposed prism minimality when confronted with this square drawing and the exact chromatic numbers. Lean 4.19.0 checks the package using `Std`, with no admitted declarations, custom axioms, or `native_decide`.